

Hamro Hospital

Professional HIS Improvement Roadmap

Complete practical list of non-UI improvements needed to make the project feasible, professional, standard, and ready for real hospital use.

Scope

This report excludes UI redesign. It focuses on features, workflows, backend architecture, permissions, finance, clinical safety, audit, and production readiness. Receipt/print standardization is included because it is operational, legal, and barcode-related.

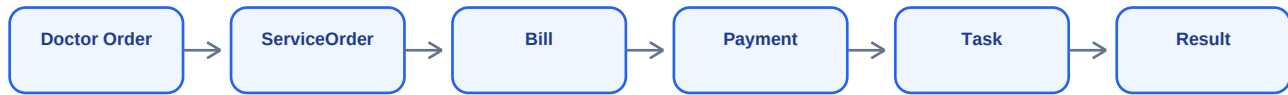
Main conclusion

The most important missing foundation is a Central ServiceOrder engine. It should connect doctor orders, referrals, billing, payments, department tasks, results, patient timeline, and finance reports.

1. Critical Must-Do Improvements

1.1 Central ServiceOrder System

Current problem: referrals, bills, lab requests, radiology requests, pharmacy sales, OT records, admission records, and blood bank requests are connected, but not under one universal order engine.



ServiceOrder field	Purpose
patient	The patient receiving the service.
ordered_by	Doctor or staff who requested the service.
source_department	Where request came from.
destination_department	Where request should be processed.
service_type/service_name	Lab, Radiology, Pharmacy, OT, Admission, Blood Bank, Nursing, etc.
quantity/unit_price/total	Billing calculation.
status/payment_status	Clinical and payment lifecycle.
referral/bill	Links to clinical referral and payment record.
created_at/updated_at/completed_at	Audit and workflow tracking.

Suggested statuses: Requested, Accepted, Pending Payment, Paid, In Progress, Completed, Cancelled, Rejected.

1.2 Central Patient Timeline

Create a PatientTimeline model that records registration, OPD visits, consultations, referrals, bills, payments, lab/radiology results, admissions, pharmacy, documents, and discharge events in one chronological view.

1.3 Real-Time Notifications

Use Django Channels and WebSockets so departments see new referrals and payment completion immediately, without refresh.

1.4 Full Pending Payment Engine

Every billable item should follow: Requested -> Pending Bill -> Payment Pending -> Paid -> Department can process.

2. Billing and Receipt Improvements

2.1 Unified Receipt Template

Create templates/includes/standard_receipt.html and reuse it for OPD, billing, admission, pharmacy, laboratory, radiology, OT, blood bank, and insurance prints.

Required field	Reason
Hospital logo/name/address/phone	Professional identity.
Patient name/ID/phone	Patient identification.
Patient barcode	Mandatory scan-based workflow.
Receipt or bill number	Finance tracking.
Department/service/doctor	Operational clarity.
Date/time/status/payment method	Audit and payment tracking.
Amount	Financial record.
Printed by/printed date	Legal and audit trail.
Bill/module barcode	Fast cashier/department lookup.

2.2 More Detailed Bill Status

Recommended statuses: Draft, Pending Payment, Partially Paid, Paid, Insurance Pending, Insurance Approved, Refund Requested, Refunded, Cancelled.

2.3 PaymentEvent Model

A bill can have multiple payments: cash, eSewa, insurance, card, partial payment, advance deposit. PaymentEvent should store amount, method, received_by, transaction_reference, gateway_response, paid_at, and remarks.

2.4 eSewa for Pending Bills

Pending bills should support Pay with eSewa. After success, bill becomes Paid, department is notified, Finance updates, and audit log is written.

3. Doctor Module Improvements

3.1 Doctor Clinical Workspace

Doctor dashboard should show today's patients, pending consultations, completed consultations, returned lab/radiology results, pending referrals, and notifications.

Recommended doctor flow: Open patient -> Write clinical note -> Add diagnosis -> Prescribe medicine -> Order lab/radiology -> Refer department -> Follow result.

3.2 Result Notifications

When lab or radiology completes a result, the requesting doctor should receive a notification such as: CBC result for P-000245 is ready.

3.3 Prescription-to-Pharmacy

Doctor prescriptions should automatically appear in Pharmacy's pending prescription queue. Pharmacy checks stock, generates sale/bill, dispenses medicine, and updates patient history.

Doctor feature	Professional requirement
Clinical note	Structured SOAP/diagnosis entry.
Orders	Create ServiceOrders for lab/radio/pharmacy.
Results	Notify doctor when completed.
Profile	Professional/contact/eSewa optional details.
Permissions	View records; no unauthorized download/delete.

4. Referral System Improvements

4.1 Referral Should Create ServiceOrders

A referral may contain many requested services. Example: CBC, LFT, KFT should create three ServiceOrder rows linked to one referral.

4.2 Structured Blood Bank Form

Fields needed: Blood Group, Component, Units Needed, Urgency, Crossmatch Required, Diagnosis, Transfusion Reason, Requested Date, Notes.

4.3 Structured Nursing Request Form

Options: IV Fluid, Dressing, Vitals Monitoring, Medication Administration, Discharge Preparation, Transfer Preparation, Catheter Care, Oxygen Monitoring, Wound Care, Other.

4.4 Stronger Doctor-to-Doctor Referral

Fields: Receiving Department, Receiving Doctor optional, Priority, Clinical Notes, Reason, Attachment, Status. If no doctor is chosen, notify all doctors in that department.



5. Admission Improvements

5.1 Full Admission Billing Engine

Admission billing should include admission deposit, daily bed charge, nursing charge, doctor round charge, procedure charge, medicine charge, lab/radiology charges, discharge bill, and insurance split.

5.2 Admission Deposit

Add admission deposit, deposit receipt, deposit used, and deposit refund.

5.3 Discharge Checklist

Checklist item	Required before discharge
All bills paid	No pending patient payments.
Lab/radiology complete	No critical pending reports.
Medicine charges complete	Pharmacy cleared.
Discharge summary	Prepared and approved.
Nursing clearance	Nursing ready for discharge.
Insurance clearance	Claim/coverage verified.
Bed release	Bed ready to free.

5.4 Bed Transfer Workflow

Support transfers between ward, cabin, ICU, HDU, and beds. Each transfer should update charges automatically.

6. Nursing Module Improvements

6.1 Nursing Requests to Admission

Nursing should send structured requests to Admission: Ready for Admission, Ready for Transfer, Ready for Discharge, Need Doctor Review, Need Billing Clearance, Need Insurance Clearance.

6.2 Nursing Care Plan

Add vitals schedule, medication schedule, nursing notes, shift handover, fluid input/output, pain score, fall risk, and wound dressing log.

Nursing item	Why needed
Vitals chart	Continuous patient monitoring.
Medication administration	Legal and care record.
Shift handover	Safer continuity of care.
Discharge readiness	Admission gets actionable update.
Care plan	Structured inpatient nursing workflow.

7. Laboratory Improvements

7.1 Lab Sample Workflow

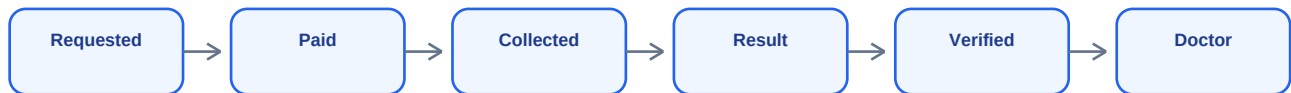
Recommended statuses: Requested, Payment Pending, Paid, Sample Collected, In Lab, Result Entered, Verified, Dispatched.

7.2 Result Verification

Add result_entered_by, verified_by, verified_at. Only verified reports should be final.

7.3 Lab Panels

Create LabPanel, LabParameter, LabResultValue. Panels like CBC, LFT, KFT, Lipid Profile and Urine Routine include multiple parameters.



8. Radiology Improvements

8.1 Radiology Workflow

Recommended statuses: Requested, Payment Pending, Paid, Scheduled, Imaging Done, Report Drafted, Verified, Delivered.

8.2 Radiology Templates

Add report templates for X-Ray Chest, USG Abdomen, CT Head, MRI Spine, Echo, ECG and other frequent imaging reports.

Radiology need	Purpose
Scheduling	Manage imaging room/time.
Report draft	Radiologist can prepare report.
Verification	Final approval before release.
Doctor notification	Doctor knows report is ready.
Templates	Faster and standardized reporting.

9. Pharmacy Improvements

9.1 Prescription Queue

Pharmacy dashboard should show Pending prescriptions, Paid prescriptions, Ready to dispense, Dispensed, and Out of stock.

9.2 Stock Validation

Before dispensing, check stock, block sale if insufficient, and suggest alternatives.

9.3 Expiry and Batch Tracking

Add batch number, expiry date, supplier, purchase price, MRP and stock by batch.

Feature	Benefit
Prescription queue	Doctor order flows directly to pharmacy.
Stock validation	Prevents negative inventory.
Batch tracking	Supports expiry and recalls.
Alternative medicine	Useful when stock is unavailable.
Supplier/purchase price	Improves pharmacy finance.

10. Blood Bank Improvements

10.1 Blood Inventory Lifecycle

Needed statuses: Collected, Tested, Available, Reserved, Issued, Expired, Discarded, Returned.

10.2 Blood Compatibility Check

System should warn if incompatible blood group is selected. Example: O negative can donate to all; AB positive can receive from all.

Blood Bank item	Purpose
Compatibility check	Patient safety.
Crossmatch workflow	Safe transfusion.
Reserve/issue status	Inventory accuracy.
Expiry/discard	Legal and safety compliance.
Blood product types	Whole blood, PRBC, plasma, platelets.

11. Insurance Improvements

11.1 Insurance Approval Workflow

Needed statuses: Claim Created, Submitted, Under Review, Approved, Partially Approved, Rejected, Settled.

11.2 Insurance Coverage Split

Bill should show total amount, insurance covered, patient payable, approved amount, rejected amount.

Insurance field	Reason
Approved amount	Shows insurer liability.
Rejected amount	Shows patient/adjustment exposure.
Patient payable	Cash Counter clarity.
Cashless flag	Different claim workflow.
Settlement status	Finance reconciliation.

12. Finance and Reports Improvements

12.1 Department Revenue Dashboard

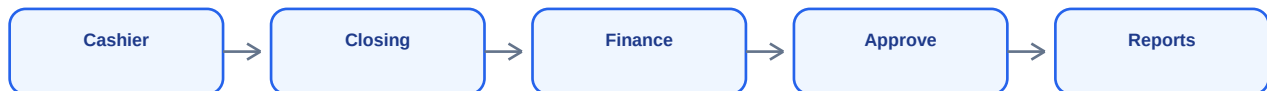
Each department should see its own revenue: Lab, Radiology, Pharmacy, Admission, OT, Blood Bank, OPD. Finance sees all.

12.2 Daily Closing

Cashier should record opening cash, total collection, cash denomination, eSewa total, card total, refund total, closing balance, and submitted-to-Finance status.

12.3 Finance Reconciliation

Finance should verify cashier submitted amount, system collection, difference, remarks, and approved by.



13. Security and Permissions

13.1 Better Button Hiding

Unauthorized users should not see buttons like Download, Delete, Admit Directly, Edit Bill unless permitted.

13.2 Object-Level Permissions

Doctor can view patient record but cannot edit demographics. Pharmacy cannot edit lab reports. Lab cannot edit billing.

13.3 Audit Log Expansion

Every important action should log: Created, Updated, Deleted, Printed, Downloaded, Paid, Refunded, Approved, Rejected, Completed.

Security area	Recommendation
Object permissions	Protect per model and per patient/department.
Button visibility	Hide disallowed actions in templates.
Audit log	Write every sensitive event.
Download controls	Only allowed roles can download.
Tests	Add permission regression tests.

14. Patient Portal and Technical Improvements

Patient Portal

Patient should see own timeline, visits, bills, payments, reports, prescriptions, admissions, appointments and documents. Online payment for pending bills should be added. Profile completion should be enforced before booking.

Technical

Add automated tests, pagination everywhere, database indexes, and background jobs using Celery or Django-Q.

Technical item	Where needed
Automated tests	Registration, doctor login, referral, pending bill, payment, lab result, discharge block.
Pagination	Patients, bills, referrals, lab/radiology queues, pharmacy sales, admissions, audit logs.
Indexes	patient_code, phone, bill_number, receipt_number, referral/payment status, created_at, department.
Background jobs	Backups, SMS, email, reports, discharge packages, daily closing.

15. Final Best Roadmap

Phase	Work	Expected result
Phase 1	ServiceOrder, PaymentEvent, PatientTimeline, Unified Notification	Core HIS backbone.
Phase 2	Department queues for Lab, Radiology, Pharmacy, Blood Bank, Nursing, OT	Consistent department workflow.
Phase 3	Pending bill engine, eSewa pending bill payment, daily closing, reconciliation	Professional finance workflow.
Phase 4	Admission deposit, bed transfer, charge automation, discharge checklist	Real inpatient management.
Phase 5	WebSockets, pagination, audit hardening, tests, indexes, backups	Production readiness.

Honest conclusion

The project is now more connected and workable, but the professional next step is the Central ServiceOrder system. Without it, each department will continue to grow separate logic. With it, the system becomes one hospital-wide workflow: Doctor Order -> ServiceOrder -> Pending Bill -> Payment -> Department Task -> Result -> Patient Timeline -> Finance Report.